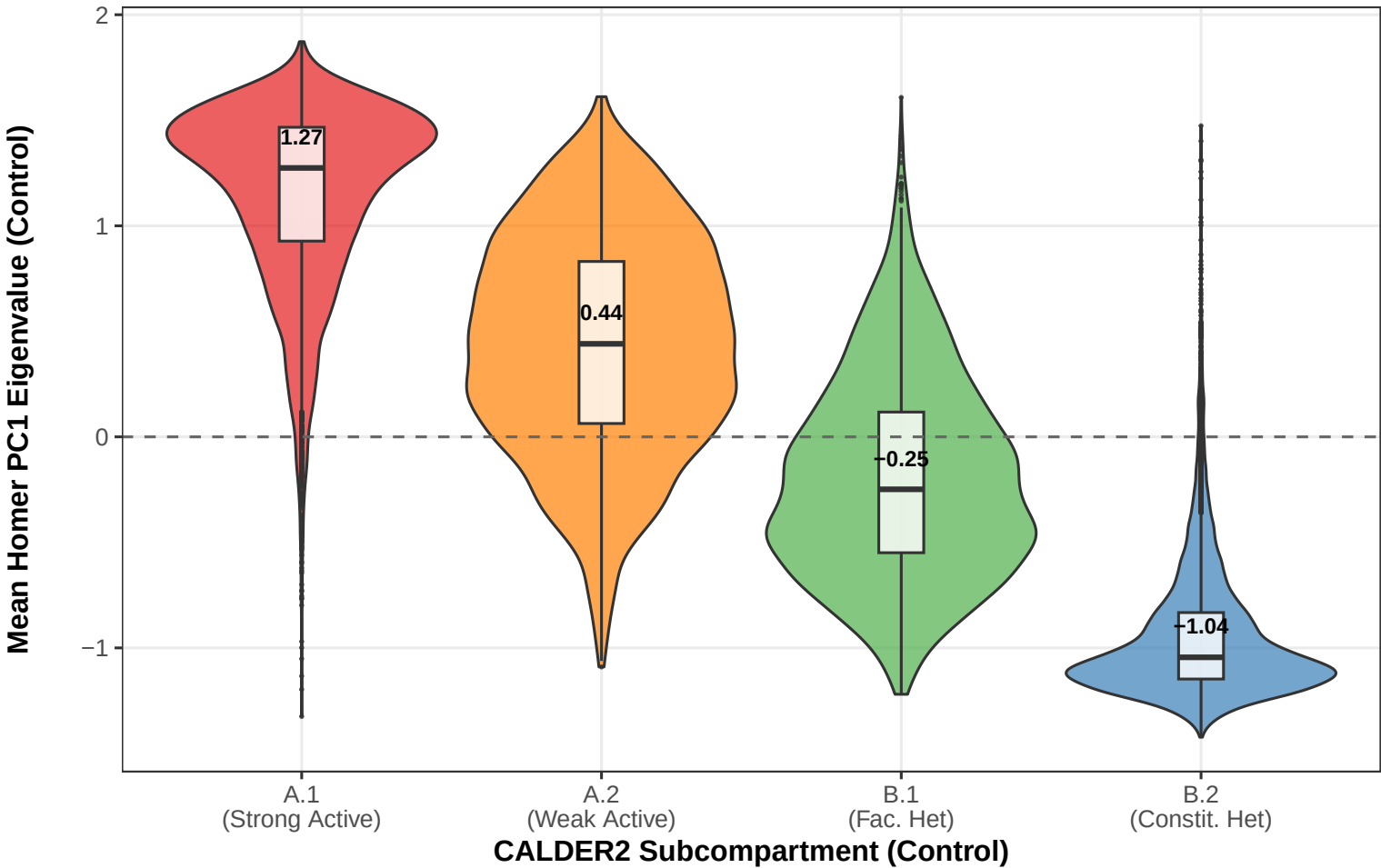


Cross-Validation: CALDER2 Subcompartments vs Homer PC1 Eigenvalues

Two independent compartment-calling methods show strong agreement

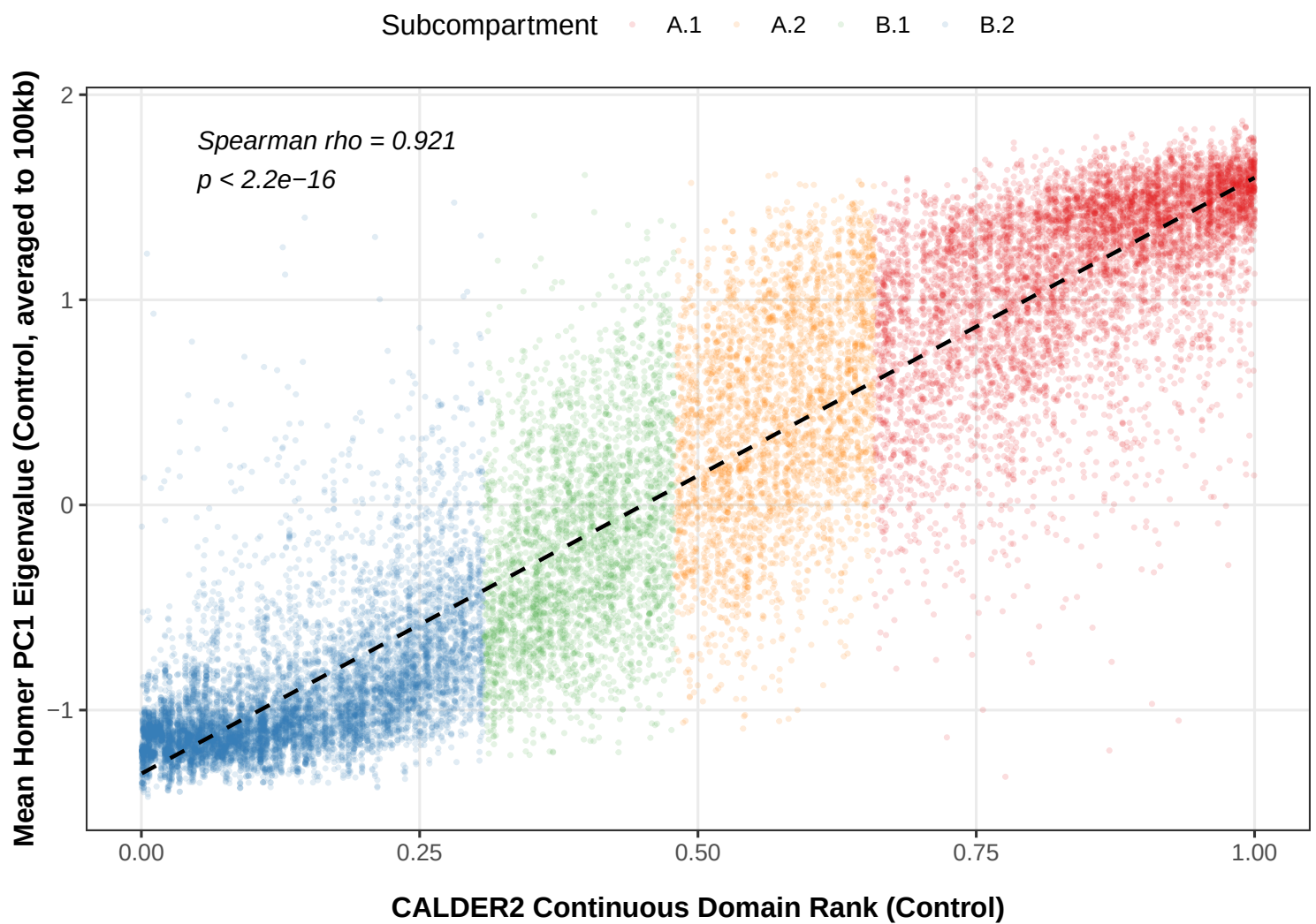
Homer PC1 Eigenvalue by CALDER2 Subcompartment

Clear ordering validates subcompartment calls | Kruskal-Wallis  $p < 2.2e-16$



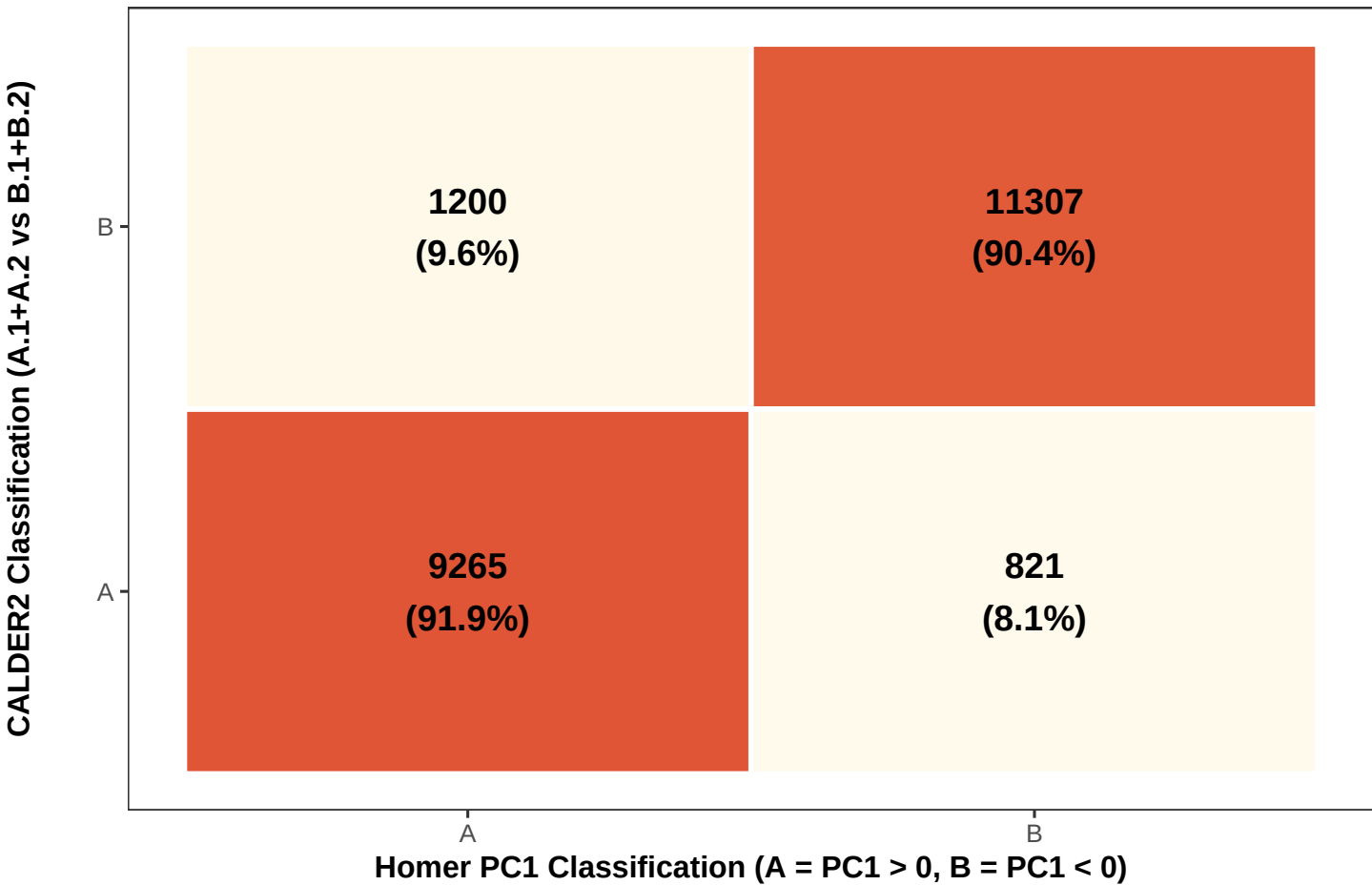
CALDER2 Continuous Rank vs Homer PC1 Eigenvalue

Two independent tools show strong agreement on compartment identity



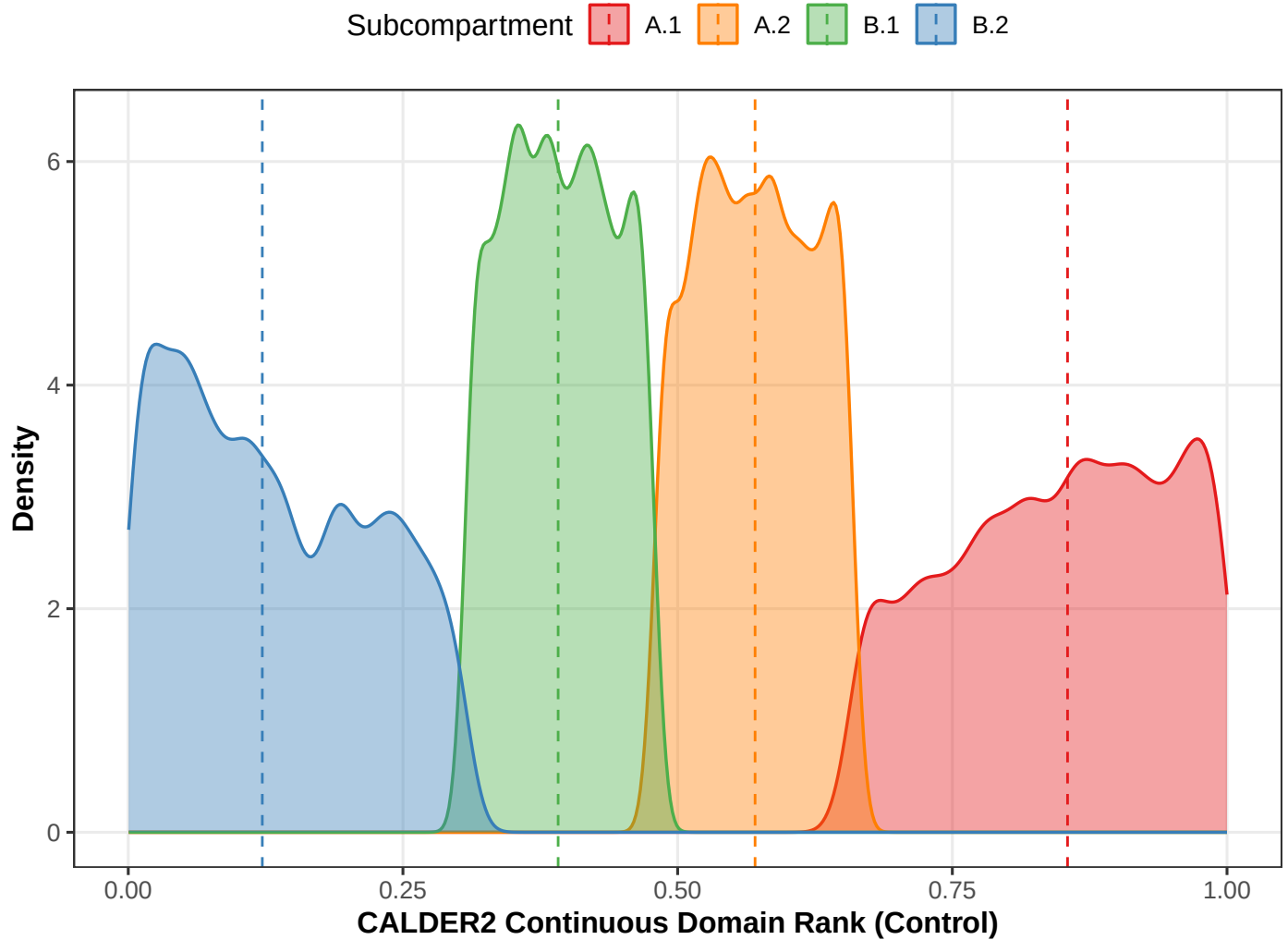
Concordance: CALDER2 vs Homer PC1 Compartment Classification

Accuracy = 91.1% | Cohen's kappa = 0.820



CALDER2 Continuous Domain Rank Distribution by Subcompartment

Rank separation validates the 4-way classification (all KS  $p < 2.2e-16$ )



Compartment Switching Concordance: CALDER2 vs Homer

Fisher's OR = 10.86,  $p < 2.2e-16$  | Do the tools agree on which bins switch?

